

Quoc (Alex) Huynh

BEng (Hons) — Electrical & Electronic Engineering | James Cook University
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Profile

Penultimate Electrical and Electronic Engineering student at James Cook University, Cairns with a strong foundation in analogue circuit design, developed through university lab work. Particularly drawn to the application of electronics in consumer devices and biomedical technology, where precision circuit design has real-world impact. Seeking an internship role where I can contribute to and grow within a capable engineering team.

Technical Skills

Technical	Software & Tools	Laboratory
Circuit analysis (KVL, KCL, Thevenin/Norton)	MATLAB (data analysis, curve fitting)	Oscilloscope & function generator
Analogue circuit design & op-amp circuits	LTSpice (circuit simulation)	Digital multimeter
NTC thermistor & sensor interfacing	Arduino IDE	Breadboard prototyping & debugging
Voltage divider & impedance matching	Desmos & ThingSpeak	Lab report writing
PCB design fundamentals	KiCad	PCB soldering & bench testing
Embedded systems (Arduino, WeMos, Raspberry Pi Pico)	Python (OOP, file I/O, Kivy GUI)	CNC milling machine stepper motors

Education

Bachelor of Engineering (Honours) — Electrical & Electronic Engineering Feb 2024 – Dec 2027
James Cook University, Cairns

- ATAR: 89.4 (Queensland Certificate of Education, Smithfield State High School, 2022)
- Relevant coursework: Electric Circuits (EG1012), Circuit Design (EE2300), Electronic Applications (EE3300)
- Key project: Additive Audio Mixer — designed, simulated in LTSpice, laid out in KiCad, and had fabricated via JLCPCB; demonstrated live at assessment driving a 16Ω speaker load (EE2300)

Selected Projects

NTC Thermistor Temperature Sensing System

Semester 2, 2024

EG1012 Electric Circuits — James Cook University

- Calibrated an NTC thermistor across a 10–70°C range and validated against the Steinhart-Hart equation using MATLAB
- Designed an optimal voltage divider to maximise ADC input voltage range across 0–100°C operating window
- Performed Thevenin equivalent analysis and designed an LM258 op-amp voltage follower to resolve impedance mismatch

- Assembled and debugged the full circuit, integrated Arduino firmware, and streamed live temperature data to ThingSpeak via API
- Achieved measurement accuracy within 1.4°C; documented and analysed sources of error including component tolerances

CNC Milling Machine Control System

Trimester 1, 2025

CC2511 Digital Logic and Embedded Systems — James Cook University

- Wrote a modular C program for a Raspberry Pi Pico RP2040 implementing 10-direction manual control, coordinate-based movement, return-to-home, PWM spindle with soft-start, and real-time emergency stop
- Diagnosed and resolved hardware faults including a faulty CNC shield, blown fuses, and an emergency-stop bug; successfully demonstrated all base requirements at assessment

Additive Audio Mixer – Design, Simulation & PCB Fabrication

Trimester 3, 2025

EE2300 Circuit Design – James Cook University

- Designed a three-stage mixer (BJT common-emitter pre-amp, JFET summing stage, BJT emitter-follower buffer) through hand calculations verified in LTSpice before committing to KiCad PCB layout; had the board fabricated via JLCPCB and bench-tested against simulated results
- Identified and resolved output clipping through iterative LTSpice tuning; demonstrated a fully functional system driving a live 16Ω speaker load at assessment

5 MHz Function Generator – Colpitts Oscillator

Block 2, 2026

EE3300 Electronic Applications – James Cook University

- Adapted a single-supply Colpitts oscillator, selected an LC low-pass filter over RC using FFT analysis in LTSpice for superior harmonic rejection, and implemented adjustable amplitude control via an AD8061 inverting op-amp with a variable feedback resistor
- Integrated all five stages into a KiCad schematic and PCB layout with a 3D model, had the board fabricated via JLCPCB, and delivered a working 5 MHz function generator with sine and square wave outputs

Work Experience

Shift Manager

June 2019 – Present

McDonald's, Smithfield — Cairns

- Lead shifts of up to 20 crew members, coordinating task allocation and maintaining service standards under pressure
- Resolve customer and operational issues quickly and effectively, developing strong problem-solving instincts
- Train and mentor new team members, building communication and leadership skills directly transferable to engineering team environments
- Manage time-critical workflows across multiple stations simultaneously, demonstrating strong time management and prioritization

Educator

June 2023 – November 2023

School Plus OSHC, Nundah

- Designed and facilitated structured activity clubs, adapting delivery to suit different learning styles across Year 1–6 age groups
- Developed patience, clear communication, and the ability to explain concepts accessibly — skills valuable in technical documentation and client-facing engineering roles

References

- Tamara Brown: Tamara.brown@au.stores.mcd.com
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